KAPLAN

Introduction to Data Communications

Topic 06 – *Providing an Overview of Wireless Networks, Technologies, and Medium Access*

Our vision is to be Singapore's private education institution of choice, and we are committed to helping our students achieve their career and educational goals through lifelong learning opportunities.

We build futures, one success story at a time.

Integrity . Knowledge . Support . Opportunity . Results



Topic 6 – Learning Outcomes

- **Wi-Fi Technology**
- **Last Mile Technologies - Cellular Networks**

KAPLAN

Wi-Fi Introduction

1. Wi-Fi stands for Wireless Fidelity, a technology that allows devices to connect to the internet or communicate with each other wirelessly within a certain range.
2. It operates on radio frequencies, typically 2.4 GHz or 5 GHz bands, and enables data transmission between devices via wireless routers or access points.

KAPLAN

Access Point (AP)

- An access point is a networking hardware device that allows Wi-Fi-enabled devices to connect to a wired network.

- It acts as a bridge between wireless devices and a wired network infrastructure, providing wireless connectivity to multiple devices simultaneously.

KAPLAN

Mobile Hotspot

- A mobile hotspot is a feature available on smartphones and some other devices that enables them to function as portable routers or access points.
- It allows other devices to connect to the internet using the mobile device's cellular data connection, effectively creating a Wi-Fi network on the go.

Portable Hotspot

- A portable hotspot is similar to a mobile hotspot but can also refer to standalone devices, such as portable Wi-Fi routers or dongles, that provide internet access via cellular networks.

- These devices are not tied to a specific smartphone and can be used with any compatible device that can connect to Wi-Fi.

KAPLAN

Tethered Connection

- A tethered connection refers to the process of connecting a device to another device, typically via a physical cable like USB, to share internet connectivity or resources.

- For example, tethering a smartphone to a laptop via USB allows the laptop to access the internet through the smartphone's cellular data connection.

KAPLAN

Untethered Connection

- An untethered connection refers to wireless connections where devices are not physically connected to each other.

- Wi-Fi and Bluetooth are examples of untethered connections, allowing devices to communicate and share data without the need for physical cables.

KAPLAN

Wi-Fi Modes

- **Ad hoc mode**: In ad hoc mode, Wi-Fi devices communicate directly with each other without the need for a central access point (AP). It's often used for peer-to-peer connections between devices.

- **Infrastructure mode**: In infrastructure mode, Wi-Fi devices communicate through a central access point (AP). This mode is commonly used in homes, offices, and public Wi-Fi networks.

KAPLAN

Wi-Fi Terms

- **BSS (Basic Service Set)**: A BSS is a group of stations that communicate with each other. It can be a standalone AP or a group of stations associated with a single AP.

- **BSSID (Basic Service Set Identifier)**: A BSSID is a unique identifier assigned to each BSS. It's typically the MAC address of the AP in infrastructure mode or the MAC address of the device in ad hoc mode.

- **SSID (Service Set Identifier)**: An SSID is a name used to identify a wireless network. It's what users see when scanning for available Wi-Fi networks.

- **ESS (Extended Service Set)**: An ESS is a set of interconnected BSSs. It allows devices to roam between APs within the same network seamlessly.

KAPLAN

Wi-Fi Scanning Methods

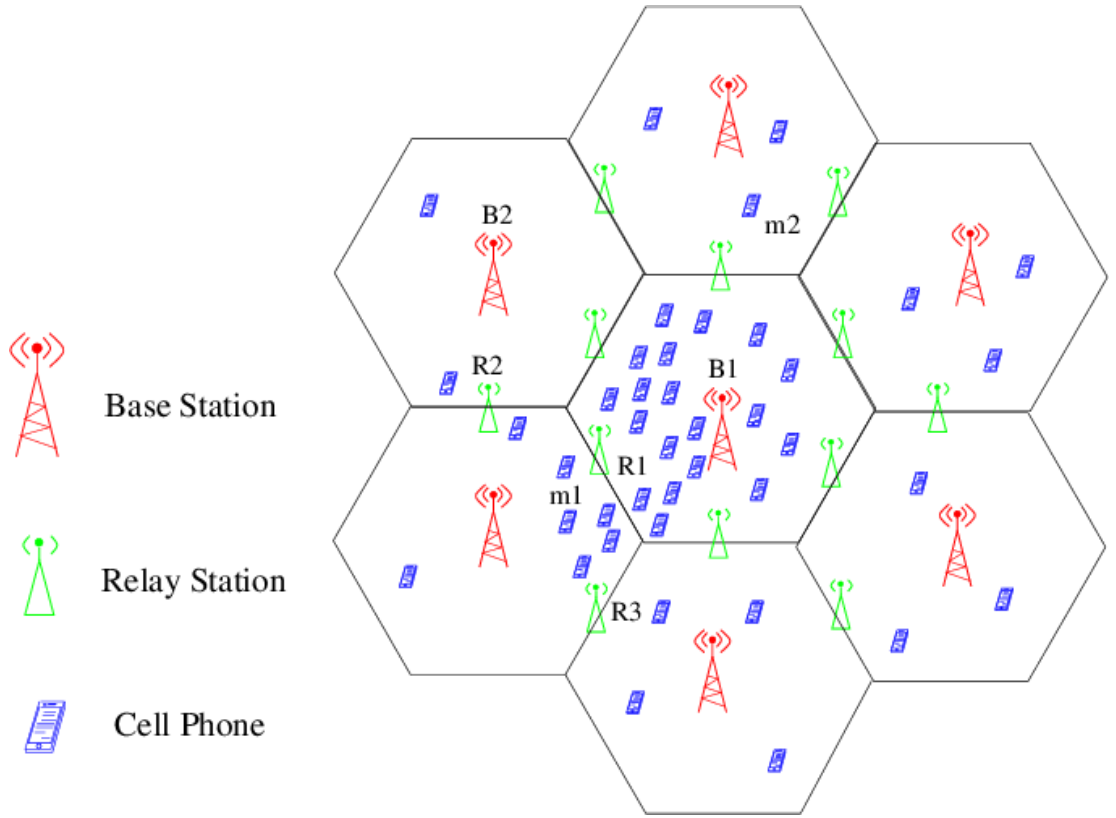
- **Active scanning**: In active scanning, a Wi-Fi device sends out probe requests to discover nearby networks. Access points respond with probe responses containing information such as SSID and capabilities.

- **Passive scanning**: In passive scanning, a Wi-Fi device listens for beacon frames transmitted by nearby access points. Beacon frames are periodically broadcasted by APs to announce their presence and network information.

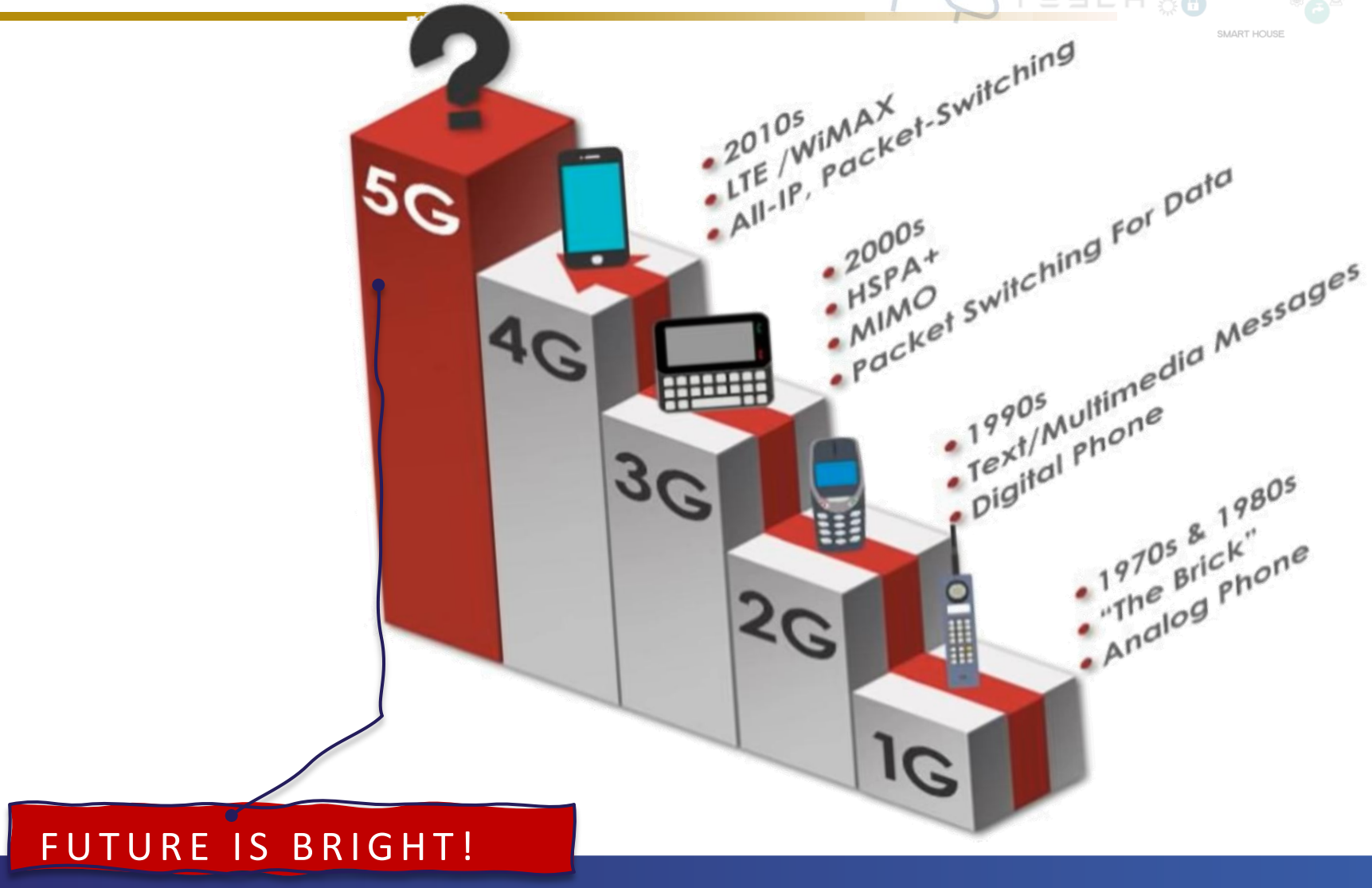
KAPLAN

Cellular Networks

- High-speed, high-capacity voice and data communication networks
- Enhanced multimedia and seamless roaming capabilities for supporting cellular devices.



Cellular Networks



Cellular Networks



- **1G**: Analog signals only: voice calls
- **2G**: Ran on digital signals. SMS (and MMS).
- **3G**: Still in use today. Greater voice and data capacity and speed. Mobile phones are no longer just about making calls, but hubs of social connectivity.
- **4G**: Much faster than 3G (Up to 500Mbps). Offer connectivity for tablets and laptops as well as smartphones.

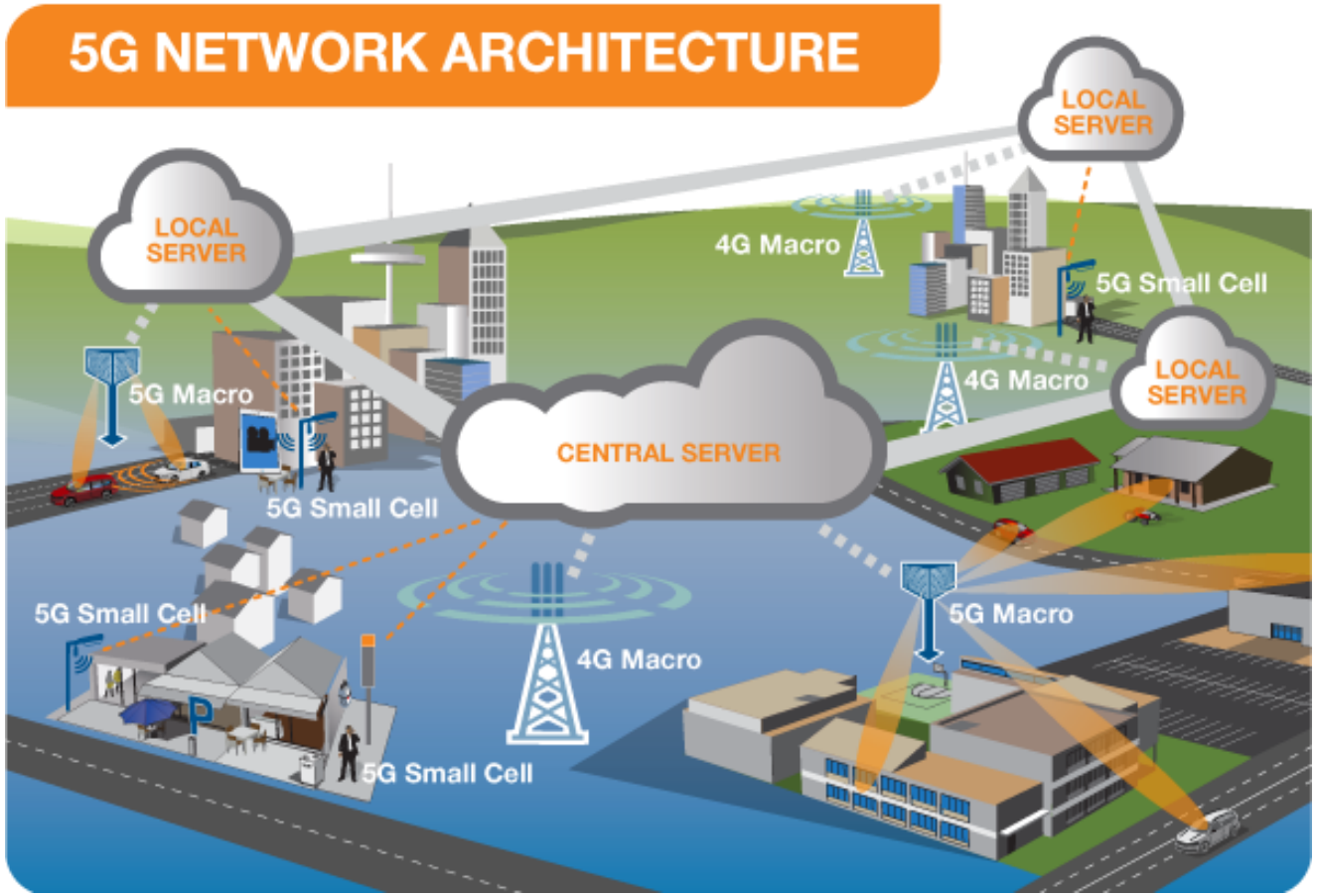
Cellular Networks

- **2020s: 5G era**
- High speed, low latency, and high connectivity
- 3 major categories:
  1. **Machine-to-machine communication: IoT**
  2. **Ultra-reliable low latency communications**
  3. **Enhanced mobile broadband**



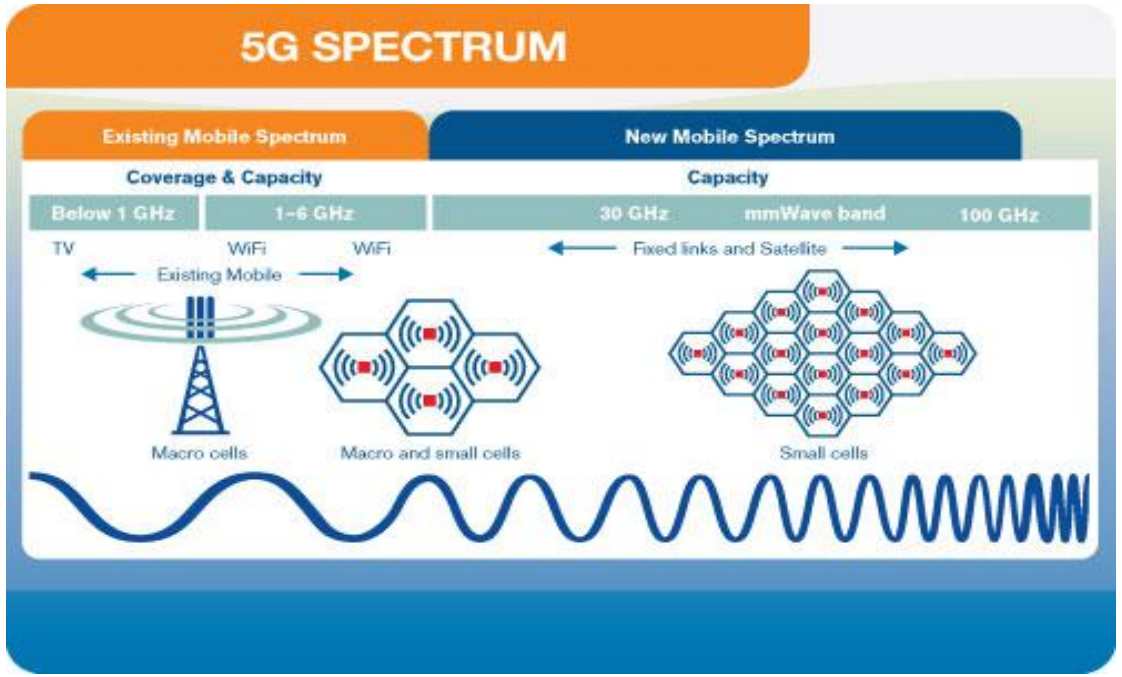
5G Technologies

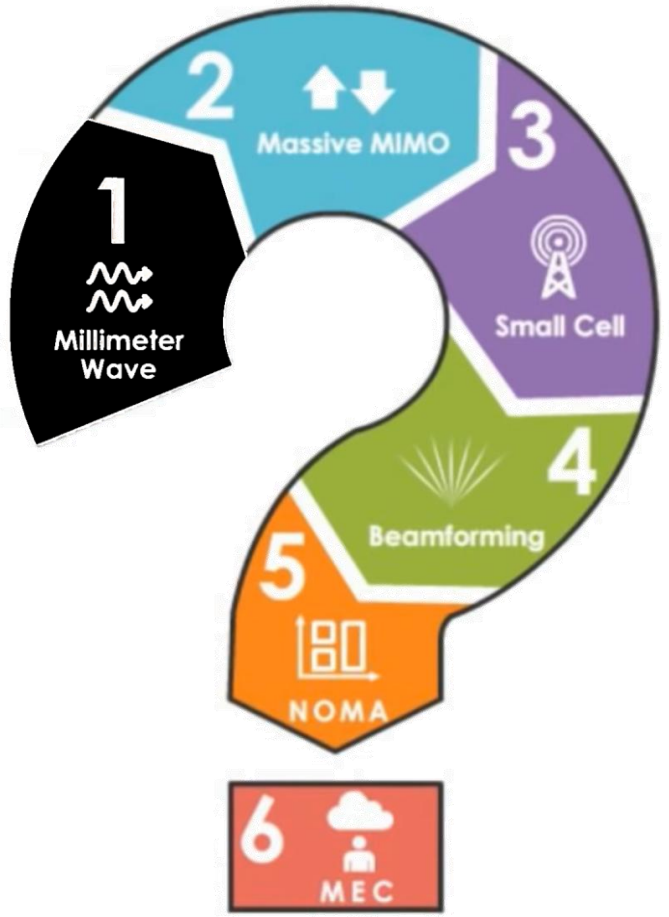
1. Millimeter Wave
2. Massive MIMO
3. Small Cell
4. Beamforming
5. NOMA
6. MEC



Millimeter Wave

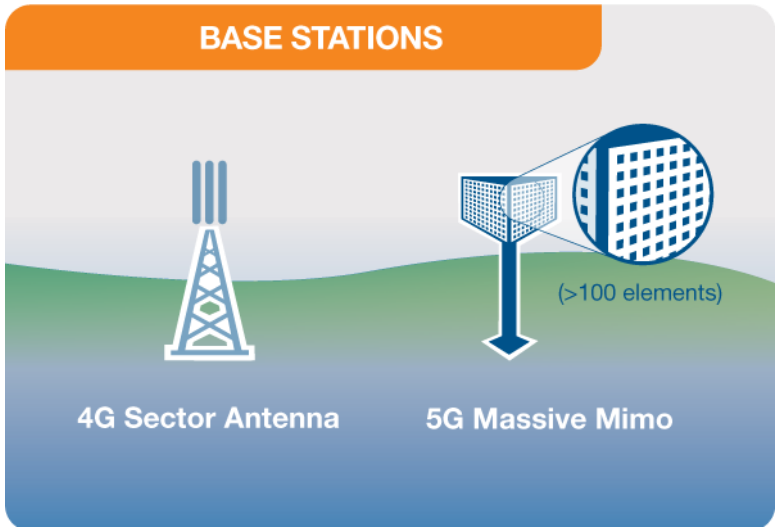
- New and less used band
- Higher frequency wave carry more data
- Supports having a massive MIMO antenna



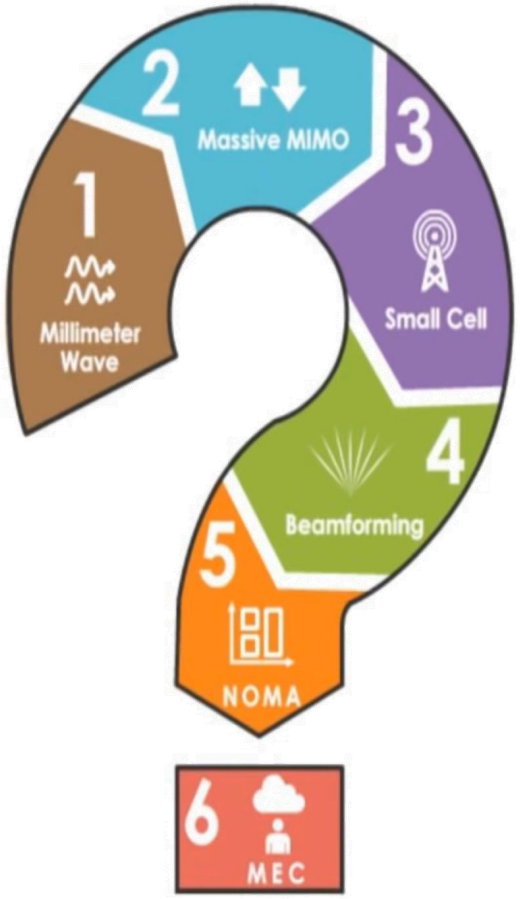


Massive MIMO

- **MIMO**: Multiple Input Multiple Output
- A lot of transmitters and receivers installed on a small size cell or panel

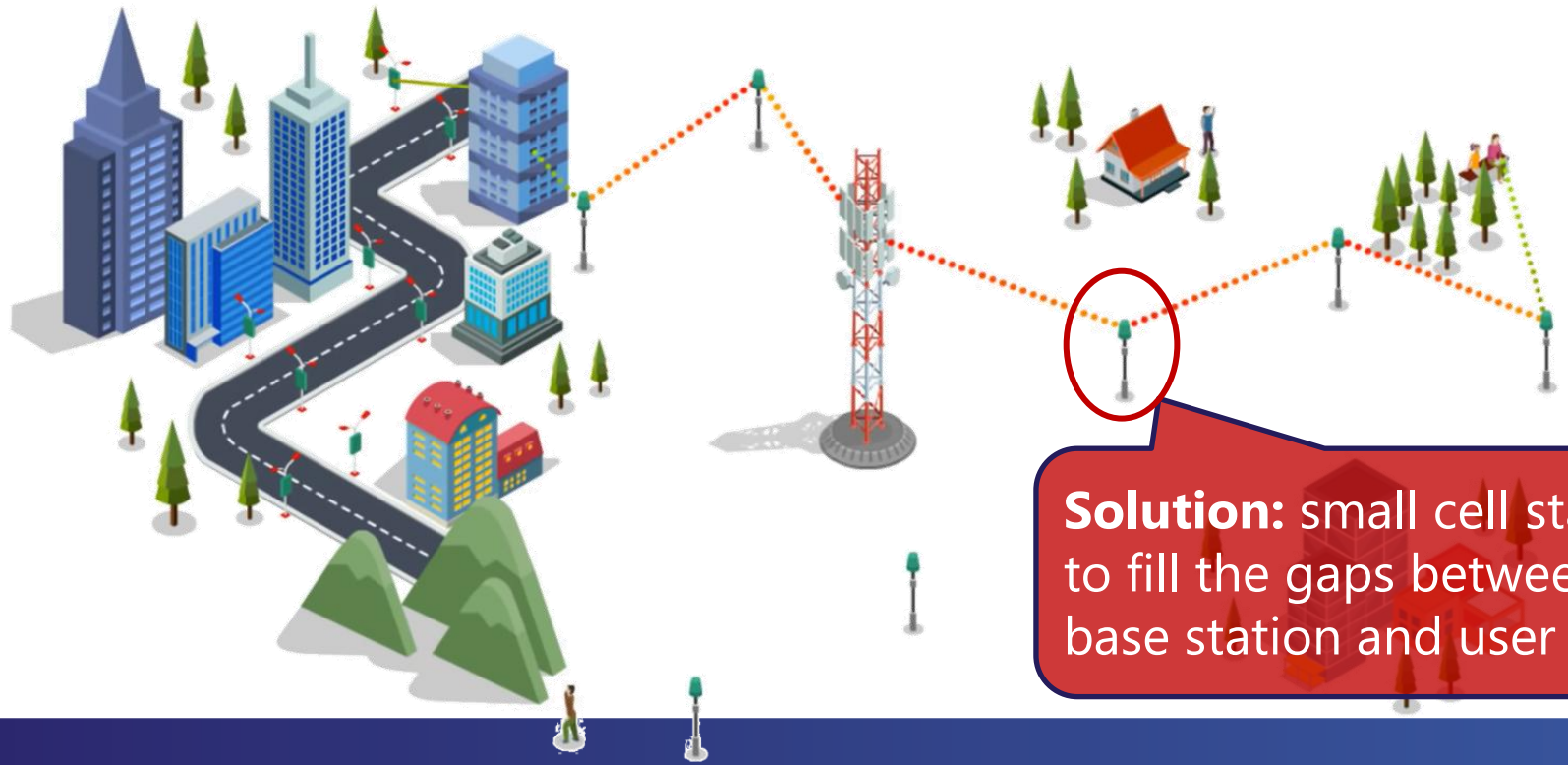




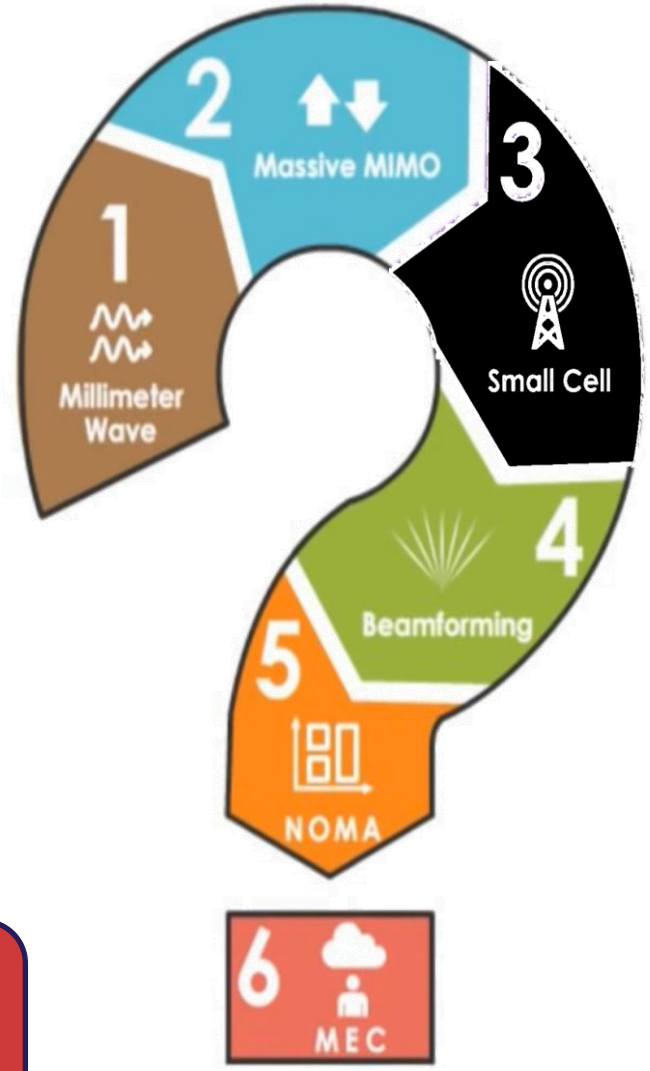


Small Cell

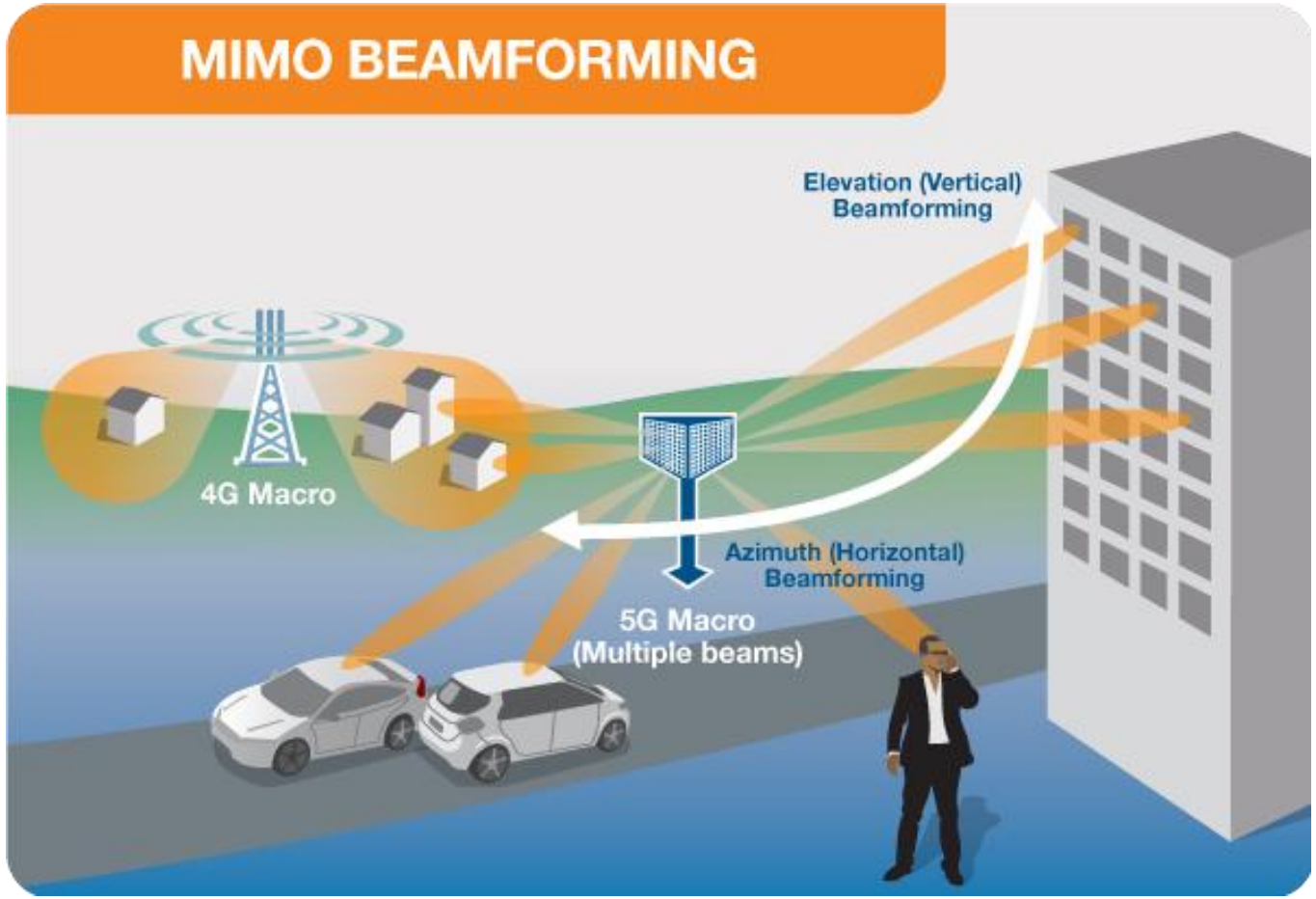
- **High frequency signals** have **more collisions** with obstacles in the air/on the ground
- Thus can only **cover shorter distance**

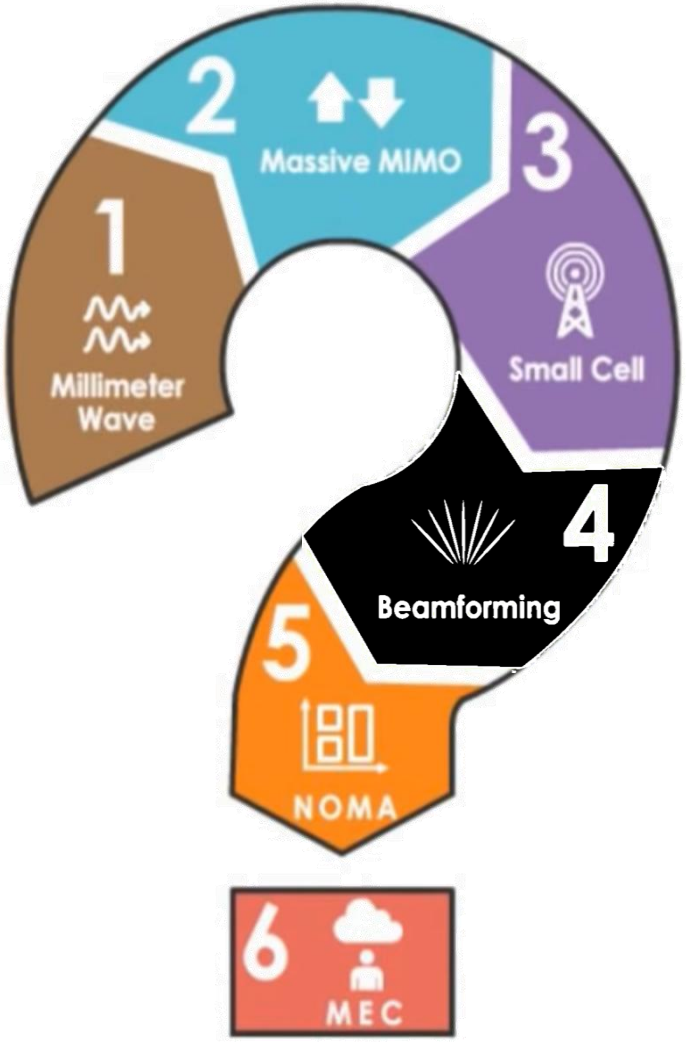


Solution: small cell stations to fill the gaps between base station and user



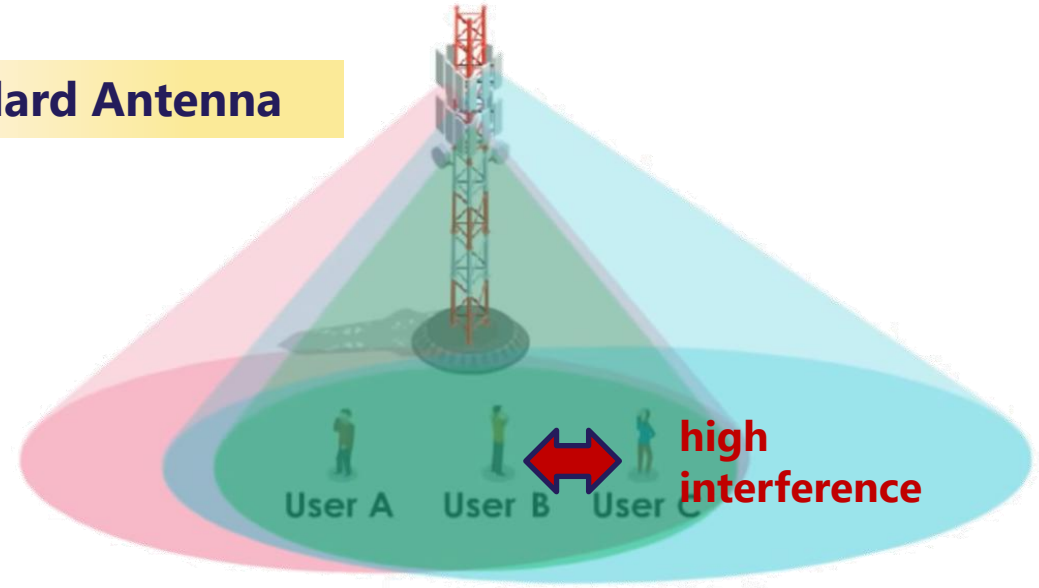
Beamforming

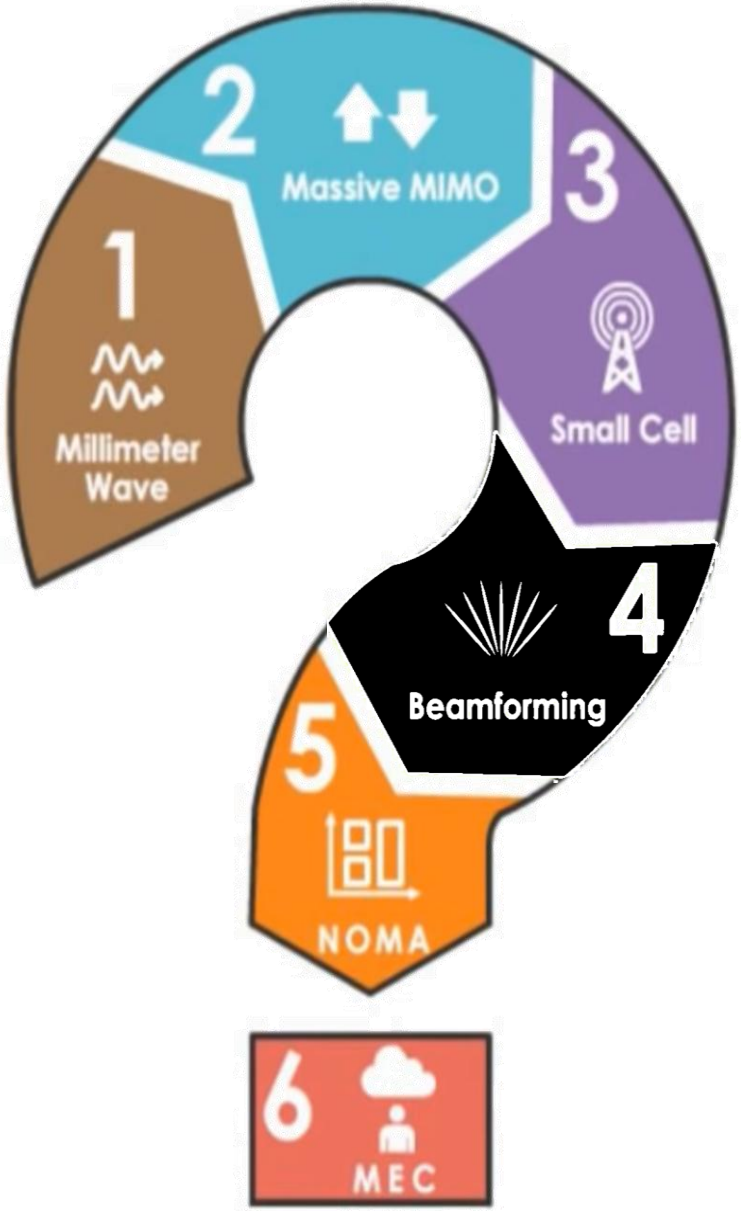




Beamforming

- **4g signals are omni directional** spreads over large area as they travel – lose energy
- **Interference is worse** when users are nearby

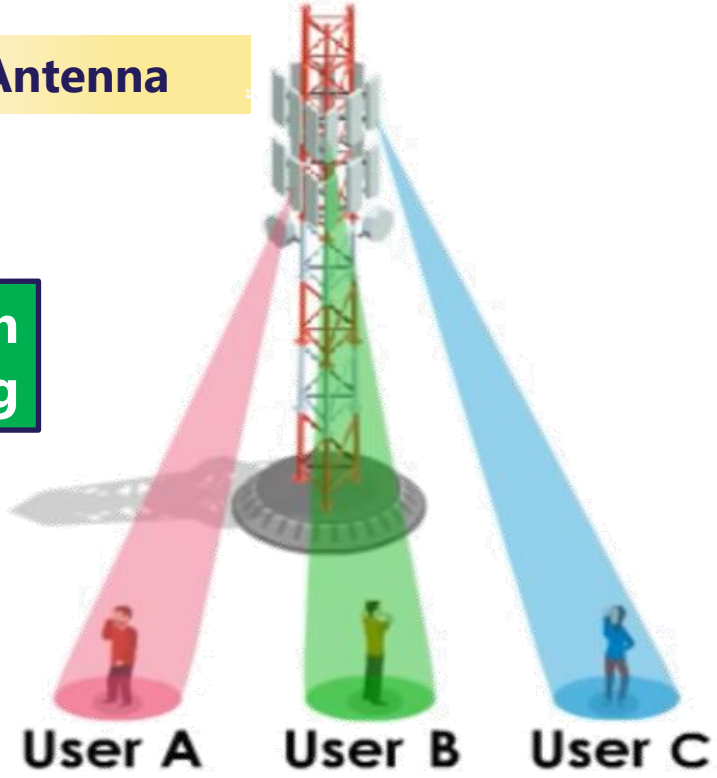




Beamforming

5G Antenna

With Beamforming

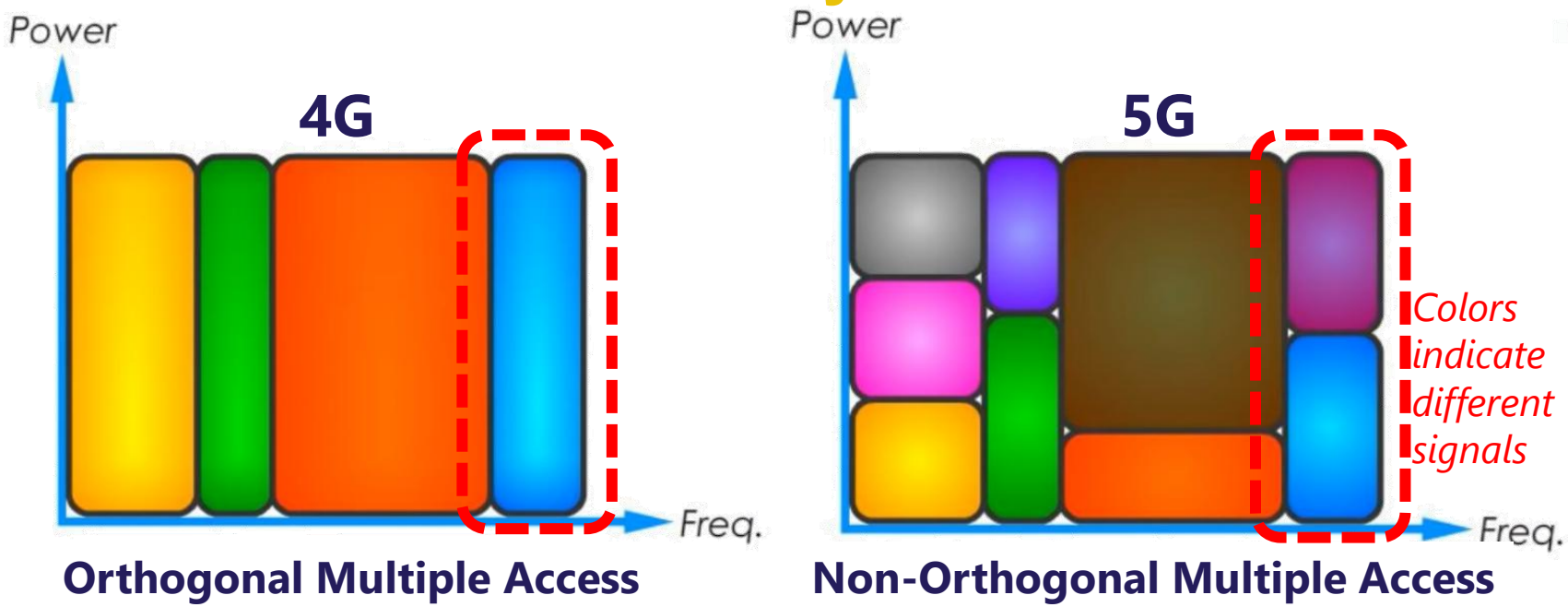




- Transmission between user and base/cell station **more directional** (i.e. laser beam)
- Less interference, less energy consumption

NOMA

- Non-Orthogonal Multiple Access
- Allow **different signals share** the **same channel simultaneously**

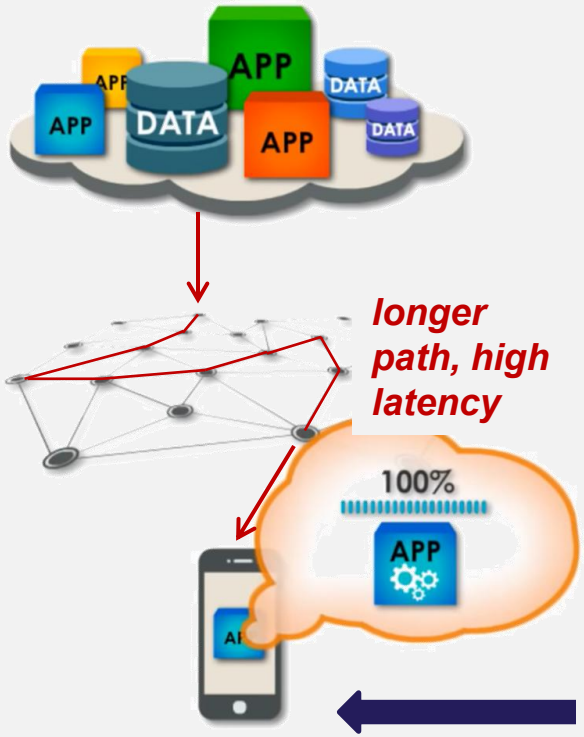




Mobile Edge Computing (MEC)

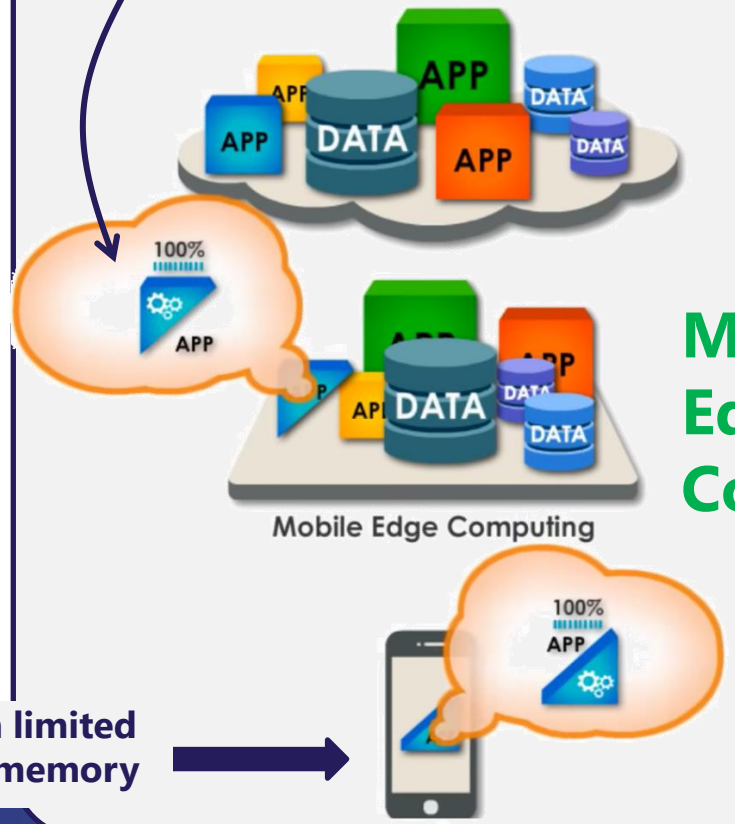
Cloud Computing

- Cloud is far away from users
- Users required to download many data and apps

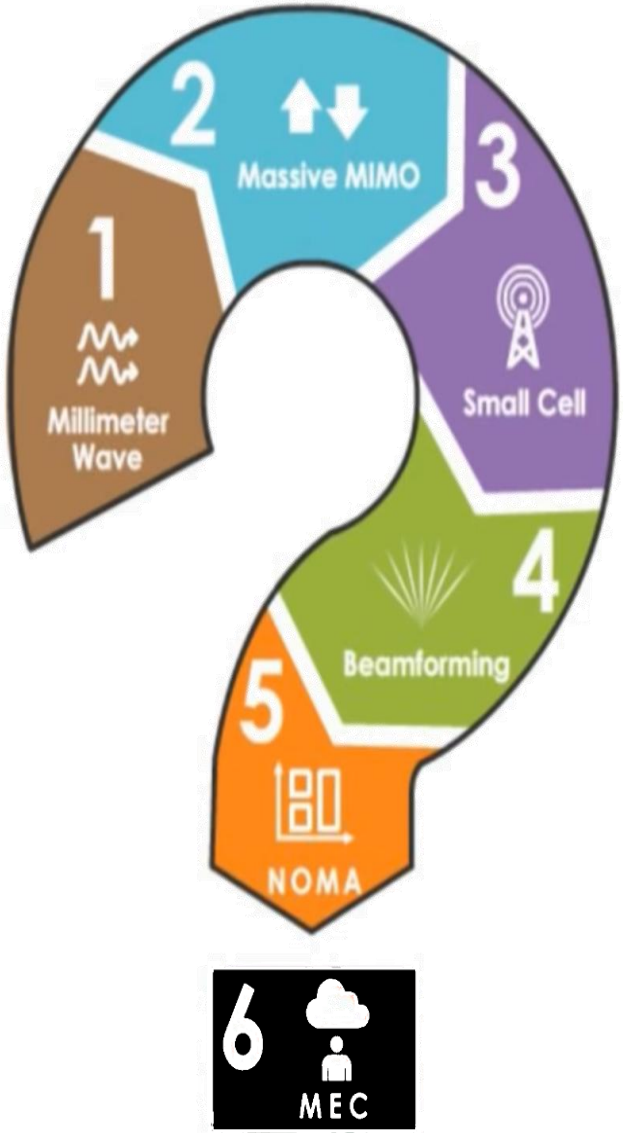


Move cloud computing and **services** closer to the device (**edge of the** local **network**)

Application splitting (some tasks performed at device and some at the cloud)



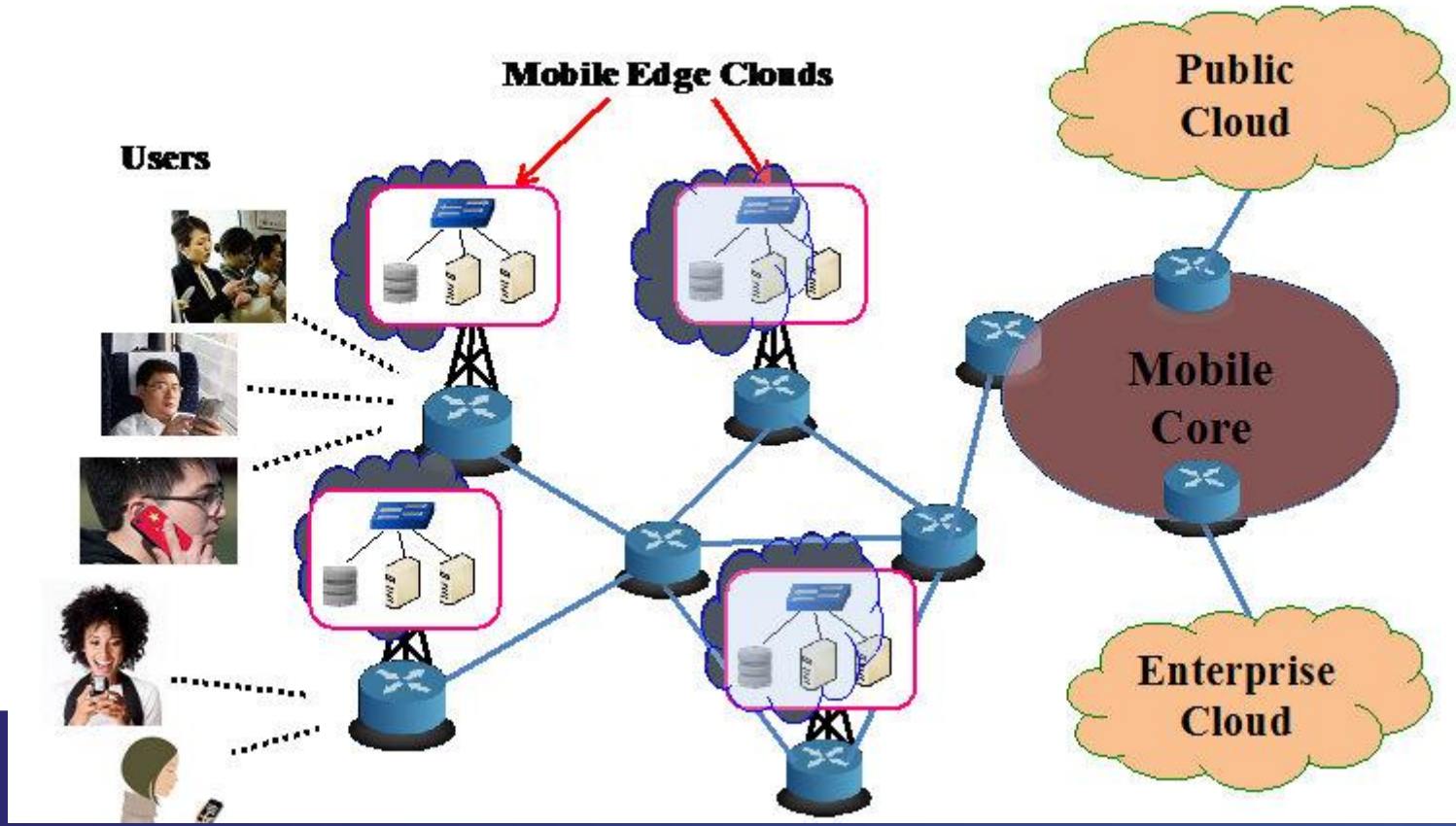
Mobile Edge Computing

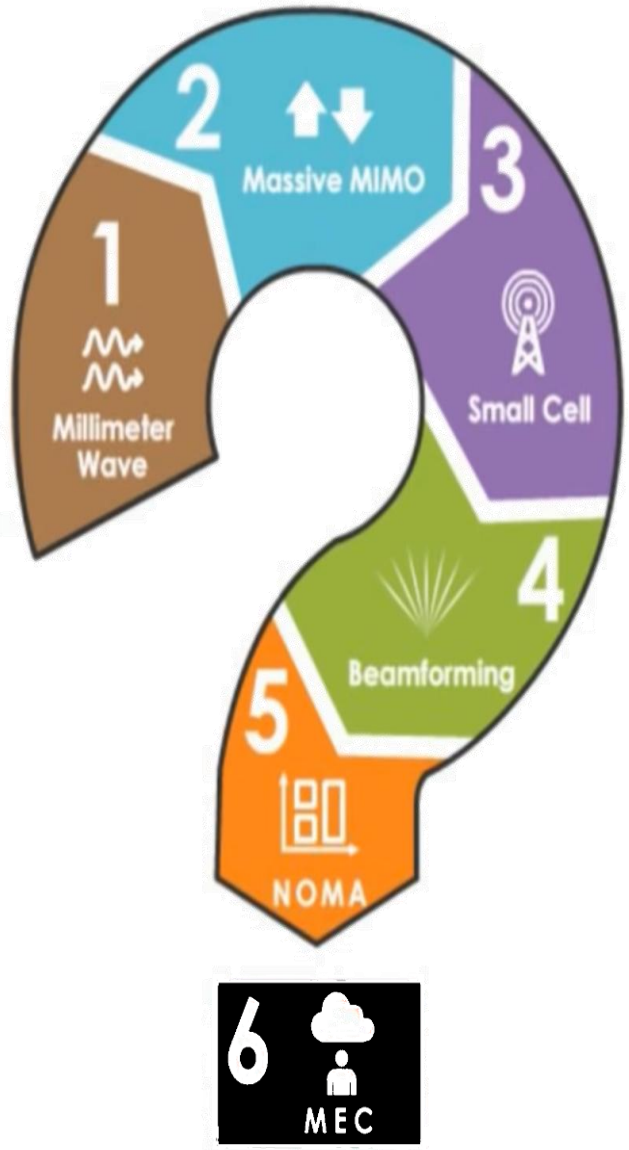


Mobile Edge Computing (MEC)

computing, storage and networking resources are integrated with the base station

Compute intensive and latency sensitive applications like augmented reality and image processing can be hosted at the edge of the network





IEEE 802.11 Standards

1. **802.11b**: Introduced in 1999, 802.11b operates in the 2.4 GHz band and offers a maximum data rate of 11 Mbps.
2. **802.11a**: Introduced in 1999, 802.11a operates in the 5 GHz band and offers a maximum data rate of 54 Mbps.
3. **802.11g**: Introduced in 2003, 802.11g operates in the 2.4 GHz band and offers a maximum data rate of 54 Mbps.
4. **802.11n**: Introduced in 2009, 802.11n operates in both the 2.4 GHz and 5 GHz bands and offers higher data rates, up to 600 Mbps.
5. **802.11ac**: Introduced in 2013, 802.11ac operates in the 5 GHz band and offers even higher data rates, up to several gigabits per second.
6. **802.11ax (Wi-Fi 6)**: Introduced in 2019, 802.11ax operates in both the 2.4 GHz and 5 GHz bands and focuses on improving efficiency in high-density environments, offering higher data rates and better performance.

KAPLAN

MIMO (Multiple Input Multiple Output)

- MIMO technology uses multiple antennas at both the transmitter and receiver to improve communication performance by transmitting multiple data streams simultaneously.

- It enhances data throughput, range, and reliability by exploiting multipath propagation and spatial diversity.

KAPLAN

IEEE 802.11 - CSMA/CA

1. CSMA/CA stands for Carrier Sense Multiple Access with Collision Avoidance.
2. It's a protocol used in wireless networks to regulate access to the shared medium (the wireless channel) and avoid collisions.
3. Before transmitting, a device listens to the channel to check if it's busy. If the channel is idle, the device waits for a random period before transmitting to avoid collisions with other devices.

KAPLAN

2.4GHz Band

- The 2.4GHz band is one of the frequency bands used for Wi-Fi communication.
- It's divided into multiple channels, typically 14 in most regions, although not all channels are available in all countries due to regulatory restrictions.
- Some channels in the 2.4GHz band overlap with each other, leading to potential interference issues, especially in densely populated areas.
- Certain channels in the 2.4GHz band, such as channels 12, 13, and 14, are restricted in some regions due to regulatory limitations.

KAPLAN

5.0GHz Band

- The 5.0GHz band, also known as the 5 GHz band, is another frequency band used for Wi-Fi communication.
- It offers more non-overlapping channels compared to the 2.4GHz band, which reduces interference and improves performance, especially in crowded environments.
- Channels in the 5.0GHz band are typically labeled as channels 36 to 165, with varying regulations depending on the country.
- Channel bonding is a technique used in the 5.0GHz band where adjacent channels are combined to increase the available bandwidth, thereby improving data throughput.

KAPLAN

OFDM (Orthogonal Frequency Division Multiplexing)

- OFDM is a modulation technique used in Wi-Fi and other wireless communication systems.
- It divides the available bandwidth into multiple orthogonal subchannels, each carrying a separate data stream.
- OFDM mitigates the effects of multipath interference and frequency-selective fading, making it suitable for high-speed data transmission over wireless channels.
- It's used in several IEEE 802.11 standards, including 802.11a, 802.11g, 802.11n, 802.11ac, and 802.11ax (Wi-Fi 6).

KAPLAN

Wireless Threat

- **Rogue AP**: A rogue access point is an unauthorized wireless access point that has been installed on a network without the knowledge or approval of the network administrator. It can be used by attackers to eavesdrop on network traffic, launch man-in-the-middle attacks, or gain unauthorized access to the network.
- **Evil Twin**: An evil twin is a rogue wireless access point that masquerades as a legitimate access point by using a deceptive SSID. When users unknowingly connect to the evil twin AP, attackers can intercept their network traffic, steal sensitive information, or deploy other malicious activities.

KAPLAN

IEEE 802.3 vs IEEE 802.11 Wi-Fi Comparison

- **IEEE 802.3 (Ethernet)**:
  - Wired standard for local area networks (LANs).
  - Uses physical cables (e.g., twisted pair, fiber optic) to transmit data.
  - Typically offers higher data rates and lower latency compared to Wi-Fi.
  - Ethernet frames are typically larger and have fixed sizes.
- **IEEE 802.11 (Wi-Fi)**:
  - Wireless standard for local area networks (LANs).
  - Uses radio waves to transmit data over the air.
  - Offers mobility and flexibility but may suffer from interference and signal attenuation.
  - Wi-Fi frames are smaller and have variable sizes due to the nature of wireless communication.

KAPLAN

IEEE 802.11 Wi-Fi Frames

- **Management Frame**: Used for network management functions such as authentication, association, and beaconing. Examples include probe request/response, association request/response, and beacon frames.
- **Control Frame**: Used for controlling the delivery of data frames. Examples include acknowledgment (ACK), request to send (RTS), and clear to send (CTS) frames.
- **Data Frame**: Carries actual data between wireless devices. Contains payload data along with header information such as source and destination addresses, frame control, and sequence control.

KAPLAN

IEEE 802.11 Wi-Fi - Data Frame

- The data frame in IEEE 802.11 Wi-Fi carries payload data between wireless devices.
- It consists of various fields including frame control, duration/ID, addresses (source and destination), sequence control, frame body (payload data), and frame check sequence (FCS) for error detection.
- Data frames are used to transmit higher-layer protocol data such as IP packets, TCP segments, or UDP datagrams over the wireless medium.

KAPLAN

IEEE 802.11 Wi-Fi vs Ethernet - Data Frame

- While both Wi-Fi and Ethernet data frames serve the same purpose of carrying payload data, they have differences in their structure and characteristics due to the underlying technologies.
- Wi-Fi data frames are smaller and have variable sizes due to the nature of wireless communication, while Ethernet frames are typically larger and have fixed sizes.
- Wi-Fi frames include additional fields such as duration/ID for medium access control, sequence control for frame sequencing, and frame check sequence (FCS) for error detection, which are not present in Ethernet frames.

KAPLAN

Thank you

